

AI-Based Quality Risk Prediction Report

Date: 20-07-2026

Time: 03:01 PM

Prediction Summary

Prediction	FAIL
Failure Risk	99.52%
Risk Level	High Risk
Likely Defect	Moisture Related Defect
Confidence	99.34%

Input Parameters

Parameter	Value
Material	PU
Braid	None
Machine	EXT-01
Target OD (mm)	10.0
Target ID (mm)	5.0
Wall Thickness (mm)	2.5
OD Tolerance (mm)	0.15
ID Tolerance (mm)	0.15
Wall Tolerance (mm)	0.08
Vacuum (bar)	-0.6417
Screw Speed (RPM)	62.31
Zone 2 Temp (°C)	203.55
Zone 3 Temp (°C)	206.58
Melt Pressure (bar)	205.01
Puller Speed (m/min)	15.795

Material Moisture (%)	0.1851
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AI Recommendations

- Increase Zone 3 Temperature from 206.58 to 216.91 °C
- Increase Melt Pressure from 205.01 to 215.26 bar
- Increase Puller Speed from 15.79 to 16.58 m/min

Generated by: AI-Based Thermoplastic Extrusion Quality Prediction System